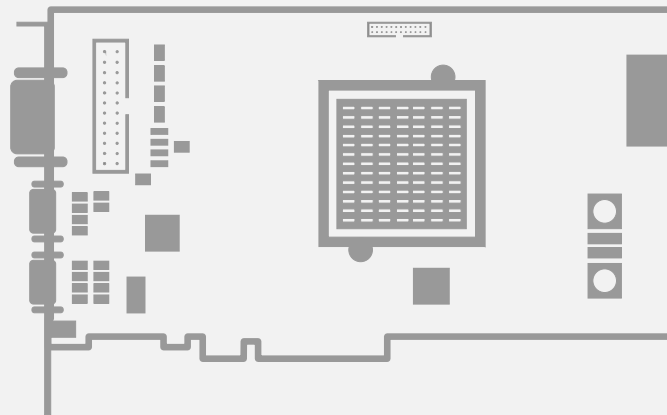


MultiCam

MultiCam Storage Formats



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1. Introduction

MultiCam frame grabbers store image pixel data into the user buffer according to a format designated by the Channel parameter **ColorFormat**.

This document provides for each format:

- The **MultiCam name**, i.e. the **ColorFormat** value
- The **PFNC name** as specified in the Pixel Format Naming Convention of GenICam
- The **storage type**:
 - **PACKED** when all the components of a pixel are stored consecutively
 - **PLANAR** when each component of a pixel is stored separately in different planes
- The **memory layout** describing how the first pixels of the first line(s) are stored in the user buffer.

2. Monochrome Pixel Formats

MultiCam Name	PFNC Name	Link
Y8	Mono8	"8-bit Monochrome" on the next page
Y10	Mono10	"10-bit Monochrome" on page 8
Y10P	Mono10p	"10-bit Monochrome lsb Packed" on page 9
N/A	Mono10pmsb	"10-bit Monochrome msb Packed" on page 10
Y12	Mono12	"12-bit Monochrome" on page 11
Y14	Mono14	"14-bit Monochrome" on page 12
Y16	Mono16	"16-bit Monochrome" on page 13

2.1. 8-bit Monochrome

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
Y8	Mono8	N/A	1 Byte/pixel

Memory Layout

@	+3	+2	+1	+0
0	Pixel[3,0]	Pixel[2,0]	Pixel[1,0]	Pixel[0,0]

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
Y10	Mono10	N/A	2 Bytes/pixel

@	+3							+2							+1							+0						
0	0	0	0	0	0	0	0	Pixel[1,0]							0	0	0	0	0	0	0	Pixel[0,0]						

2.3. 10-bit Monochrome lsb Packed

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
Y10P	Mono10p	N/A	1.25 Bytes/pixel

Memory Layout

	3 1						2 4							1 6							8							0
@	+3						+2						+1						+0									
0	... (1,0)	Pixel[2,0]						Pixel[1,0]						Pixel[0,0]														
4	... (3,0)	Pixel[5,0]						Pixel[4,0]						Pixel[3,0] (9:2)...														
8	... (5:0)	Pixel[8,0]						Pixel[7,0]						Pixel[6,0] (9:4)...														
1 2	Pixel[12,0] ...(7:0)						Pixel[11,0]						Pixel[10,0]						Pixel[9,0] (9:6)...									
1 6	Pixel[15,0]						Pixel[14,0]						Pixel[13,0]						 (9:8)									



NOTE

A pixel boundary is aligned to a 32-bit word boundary every 16 pixels (20 bytes)

2.4. 10-bit Monochrome msb Packed

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
N/A	Mono10pmsb	N/A	1.25 Bytes/pixel

Memory Layout

	3 1						2 4							1 6						8							0
@	+0						+1						+2						+3								
0	Pixel[0,0]						Pixel[1,0]						Pixel[2,0]						... (9:8)								
4	Pixel[3,0] ...(7:0)						Pixel[4,0]						Pixel[5,0]						Pixel[6,0] (9:6)...								
8	... (5:0)				Pixel[7,0]						Pixel[8,0]						Pixel[9,0] (9:4)...										
1 2	... (3,0)			Pixel[10,0]						Pixel[11,0]						Pixel[12,0] (9:2)...											
1 6	... (1,0)		Pixel[13,0]						Pixel[14,0]						Pixel[15,0]												



NOTE

A pixel boundary is aligned to a 32-bit word boundary every 16 pixels (20 bytes)

2.5. 12-bit Monochrome

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
Y12	Mono12	N/A	2 Bytes/pixel

Memory Layout

@	+3					+2	+1				+0
0	0	0	0	0	0	Pixel[1,0]	0	0	0	0	Pixel[0,0]

2.6. 14-bit Monochrome

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
Y14	Mono14	N/A	2 Bytes/pixel

Memory Layout

@	+3			+2	+1			+0
0	0	0	Pixel[1,0]			0	0	Pixel[0,0]

2.7. 16-bit Monochrome

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
Y16	Mono16	N/A	2 Bytes/pixel

Memory Layout

@	+3	+2	+1	+0
0	Pixel[1,0]		Pixel[0,0]	

3. Bayer CFA Pixel Formats

MultiCam Name	PFNC Name	Link
BAYER8	BayerBG8 BayerGB8 BayerGR8 BayerRG8	"8-bit Bayer CFA" on the next page
BAYER10	BayerBG10 BayerGB10 BayerGR10 BayerRG10	"10-bit Bayer CFA" on page 16
BAYER12	BayerBG12 BayerGB12 BayerGR12 BayerRG12	"12-bit Bayer CFA" on page 17
BAYER14	BayerBG14 BayerGB14 BayerGR14 BayerRG14	"14-bit Bayer CFA" on page 18
BAYER16	BayerBG16 BayerGB16 BayerGR16 BayerRG16	"16-bit Bayer CFA" on page 19

3.1. 8-bit Bayer CFA

MultiCam Name	PFNC Name	Color Filter Array	Storage Requirement
Bayer8	BayerBG8		1 Byte/pixel
	BayerGB8		
	BayerGR8		
	BayerRG8		

Memory Layout - BG CFA

@	+3	+2	+1	+0
0	Pixel[3,0]:G	Pixel[2,0]: B	Pixel[1,0]: G	Pixel[0,0]: B
:	:	:	:	:
H	Pixel[3,1]: R	Pixel[2,1]: G	Pixel[1,1]: R	Pixel[0,1]: G
:	:	:	:	:


NOTE

H = buffer pitch (in bytes)

3.2. 10-bit Bayer CFA

MultiCam Name	PFNC Name	Color Filter Array	Storage Requirement
Bayer10	BayerBG10		2 Bytes/pixel
	BayerGB10		
	BayerGR10		
	BayerRG10		

Memory Layout - BG CFA

@	+3	+2	+1	+0
0	0 0 0 0 0 0 0	Pixel[1,0]: G	0 0 0 0 0 0 0	Pixel[0,0]: B
:	:	:	:	:
H	0 0 0 0 0 0 0	Pixel[1,1]: R	0 0 0 0 0 0 0	Pixel[0,1]: G
:	:	:	:	:


NOTE

H = buffer pitch (in bytes)

3.3. 12-bit Bayer CFA

MultiCam Name	PFNC Name	Color Filter Array	Storage Requirement
Bayer12	BayerBG12		2 Bytes/pixel
	BayerGB12		
	BayerGR12		
	BayerRG12		

Memory Layout - BG CFA

@	+3					+2	+1					+0
0	0	0	0	0	0	Pixel[1,0]: G	0	0	0	0	0	Pixel[0,0]: B
:				:		:				:		:
H	0	0	0	0	0	Pixel[1,1]: R	0	0	0	0	0	Pixel[0,1]: G
:				:		:				:		:


NOTE

H = buffer pitch (in bytes)

3.4. 14-bit Bayer CFA

MultiCam Name	PFNC Name	Color Filter Array	Storage Requirement
Bayer14	BayerBG14		2 Bytes/pixel
	BayerGB14		
	BayerGR14		
	BayerRG14		

Memory Layout - BG CFA

@	+3		+2	+1		+0
0	0	0	Pixel[1,0]: G	0	0	Pixel[0,0]: B
:	:	:	:	:	:	:
H	0	0	Pixel[1,1]: R	0	0	Pixel[0,1]: G
:	:	:	:	:	:	:


NOTE

H = buffer pitch (in bytes)

3.5. 16-bit Bayer CFA

MultiCam Name	PFNC Name	Color Filter Array	Storage Requirement
Bayer16	BayerBG16		2 Bytes/pixel
	BayerGB16		
	BayerGR16		
	BayerRG16		

Memory Layout - BG CFA

@	+3	+2	+1	+0
0	Pixel[1,0]: G		Pixel[0,0]: B	
:	:	:	:	:
H	Pixel[1,1]: R		Pixel[0,1]: G	
:	:	:	:	:


NOTE

H = buffer pitch (in bytes)

4. RGB Color Pixel Formats

MultiCam Name	PFNC Name	Link
RGB15	BGR555	"5-5-5-bit BGR" on the next page
RGB16	BGR565	"5-6-5-bit BGR" on page 22
RGB24	BGR8	"8-bit BGR " on page 23
N/A	RGB8	"8-bit RGB" on page 24
RGB32	BGRa8	"8-bit BGRa" on page 25
N/A	RGBa8	"8-bit RGBa" on page 26
RGB30P	BGR10p	"10-bit BGR lsb Packed" on page 27
N/A	BGR10pmsb	"10-bit BGR msb Packed" on page 28
RGBI40P	BGRa10p	"10-bit BGRa lsb Packed" on page 29

4.1. 5-5-5-bit BGR

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
RGB15	BGR555	PACKED	2 Bytes/pixel

Memory Layout

@	+3			+2		+1		+0
0	0	Pixel[1,0]: R	Pixel[1,0]: G	Pixel[1,0]: B	0	Pixel[0,0]: R	Pixel[0,0]: G	Pixel[0,0]: B

4.2. 5-6-5-bit BGR

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
RGB16	BGR565	PACKED	2 Bytes/pixel

Memory Layout

@	+3		+2	+1		+0
0	Pixel[1,0]: R	Pixel[1,0]: G	Pixel[1,0]: B	Pixel[0,0]: R	Pixel[0,0]: G	Pixel[0,0]: B

4.3. 8-bit BGR

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
RGB24	BGR8	PACKED	3 Bytes/pixel

Memory Layout

@	+3	+2	+1	+0
0	Pixel[1,0]: B	Pixel[0,0]: R	Pixel[0,0]: G	Pixel[0,0]: B
4	Pixel[2,0]: G	Pixel[2,0]: B	Pixel[1,0]: R	Pixel[1,0]: G
8	Pixel[3,0]: R	Pixel[3,0]: G	Pixel[3,0]: B	Pixel[2,0]: R

4.4. 8-bit RGB

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
N/A	BGR8	PACKED	3 Bytes/pixel

Memory Layout

@	+3	+2	+1	+0
0	Pixel[1,0]: R	Pixel[0,0]: B	Pixel[0,0]: G	Pixel[0,0]: R
4	Pixel[2,0]: G	Pixel[2,0]: R	Pixel[1,0]: B	Pixel[1,0]: G
8	Pixel[3,0]: B	Pixel[3,0]: G	Pixel[3,0]: R	Pixel[2,0]: B

4.5. 8-bit BGRa

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
RGB32	BGRa8	PACKED	4 Bytes/pixel

Memory Layout

@	+3								+2	+1	+0
0	0	0	0	0	0	0	0	0	Pixel[0,0]: R	Pixel[0,0]: G	Pixel[0,0]: B

4.6. 8-bit RGBa

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
N/A	RGBa8	PACKED	4 Bytes/pixel

Memory Layout

@	+3								+2	+1	+0
0	0	0	0	0	0	0	0	0	Pixel[0,0]: B	Pixel[0,0]: G	Pixel[0,0]: R

4.7. 10-bit BGR lsb Packed

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
RGB30P	BGR10p	N/A	3.75 Bytes/pixel

Memory Layout

	3 1						2 4							1 6							8							0
@	+3							+2							+1							+0						
0	... (1,0)	Pixel[0,0]R						Pixel[0,0]G						Pixel[0,0]B														
4	... (3,0)	Pixel[1,0]R						Pixel[1,0]G						Pixel[1,0]B (9:2)...														
8	... (5:0)	Pixel[2,0]R						Pixel[2,0]G						Pixel[2,0]B (9:4)...														
1 2	Pixel[4,0]B ...(7:0)						Pixel[3,0]R						Pixel[3,0]G						Pixel[3,0]B (9:6)...									
1 6	Pixel[5,0]B						Pixel[4,0]R						Pixel[4,0]G						... (9:8)									
:																												
5 6	Pixel[15,0]R						Pixel[15,0]G						Pixel[15,0]B						... (9:8)									



NOTE

A pixel boundary is aligned to a 32-bit word boundary every 16 pixels (60 bytes)

4.8. 10-bit BGR msb Packed

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
N/A	BGR10pmsb	N/A	1.25 Bytes/pixel

Memory Layout

	3 1						2 4							1 6						8								0
@	+0							+1							+2							+3						
0	Pixel[0,0]B							Pixel[0,0]G							Pixel[0,0]R							... (9:8)						
4	Pixel[1,0]B ... (7:0)							Pixel[1,0]G							Pixel[1,0]R							Pixel[2,0]B (9:6)...						
8	... (5:0)			Pixel[2,0]G							Pixel[2,0]R							Pixel[3,0]B (9:4)...										
1 2	... (3,0)			Pixel[3,0]G							Pixel[3,0]R							Pixel[4,0]B (9:2)...										
1 6	... (1,0)			Pixel[4,0]G							Pixel[4,0]R							Pixel[5,0]B										
:																												
5 6	... (1,0)			Pixel[15,0]B							Pixel[15,0]G							Pixel[15,0]R										



NOTE

A pixel boundary is aligned to a 32-bit word boundary every 16 pixels (60 bytes)

4.9. 10-bit BGra lsb Packed

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
RGBI40P	BGRa10p	N/A	5 Bytes/pixel

Memory Layout

	3 1						2 4							1 6						8							0	
@	+3							+2							+1							+0						
0	... (1,0)	Pixel[0,0]R							Pixel[0,0]G							Pixel[0,0]B												
4	... (3,0)	Pixel[1,0]G							Pixel[1,0]B							Pixel[0,0]a (9:2)...												
8	... (5:0)	Pixel[2,0]B							Pixel[1,0]a							Pixel[1,0]R (9:4)...												
1 2	Pixel[3,0]B ...(7:0)							Pixel[2,0]a							Pixel[2,0]R							Pixel[2,0]G (9:6)...						
1 6	Pixel[3,0]a							Pixel[3,0]R							Pixel[3,0]G							... (9:8)						



NOTE

A pixel boundary is aligned to a 32-bit word boundary every 4 pixels (20 bytes)

5. RGB Color Planar Pixel Formats

MultiCam Name	PFNC Name	Link
RGB24PL	N/A	"8-bit RGB Planar" on the next page
RGB30PL	N/A	"10-bit RGB Planar" on page 32
RGB36PL	N/A	"12-bit RGB Planar" on page 33
RGB42PL	N/A	"14-bit RGB Planar" on page 34
RGB48PL	N/A	"16-bit RGB Planar" on page 35

5.1. 8-bit RGB Planar

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
RGB24PL	N/A	PLANAR	3 Bytes/pixel

Memory Layout - Plane 0

@	+3	+2	+1	+0
0	Pixel[3,0]: R	Pixel[2,0]: R	Pixel[1,0]: R	Pixel[0,0]: R

Memory Layout - Plane 1

@	+3	+2	+1	+0
0	Pixel[3,0]: G	Pixel[2,0]: G	Pixel[1,0]: G	Pixel[0,0]: G

Memory Layout - Plane 2

@	+3	+2	+1	+0
0	Pixel[3,0]: B	Pixel[2,0]: B	Pixel[1,0]: B	Pixel[0,0]: B

5.2. 10-bit RGB Planar

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
RGB30PL	N/A	PLANAR	6 Bytes/pixel

Memory Layout - Plane 0

@	+3							+2			+1						+0		
0	0	0	0	0	0	0	0	Pixel[1,0]: R			0	0	0	0	0	0	Pixel[0,0]: R		

Memory Layout - Plane 1

@	+3							+2			+1						+0		
0	0	0	0	0	0	0	0	Pixel[1,0]: G			0	0	0	0	0	0	Pixel[0,0]: G		

Memory Layout - Plane 2

@	+3							+2		+1						+0		
0	0	0	0	0	0	0		Pixel[1,0]: B			0	0	0	0	0	0	Pixel[0,0]: B	

5.3. 12-bit RGB Planar

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
RGB36PL	N/A	PLANAR	6 Bytes/pixel

Memory Layout - Plane 0

@	+3					+2					+1					+0				
0	0	0	0	0	0	Pixel[1,0]: R					0	0	0	0	0	Pixel[0,0]: R				

Memory Layout - Plane 1

@	+3					+2					+1					+0					
0	0	0	0	0		Pixel[1,0]: G						0	0	0	0		Pixel[0,0]: G				

Memory Layout - Plane 2

@	+3					+2					+1					+0					
0	0	0	0	0		Pixel[1,0]: B						0	0	0	0		Pixel[0,0]: B				

5.4. 14-bit RGB Planar

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
RGB42PL	N/A	PLANAR	6 Bytes/pixel

Memory Layout - Plane 0

@	+3			+2	+1			+0
0	0	0	Pixel[1,0]: R			0	0	Pixel[0,0]: R

Memory Layout - Plane 1

@	+3			+2	+1			+0
0	0	0	Pixel[1,0]: G			0	0	Pixel[0,0]: G

Memory Layout - Plane 2

@	+3			+2	+1			+0
0	0	0	Pixel[1,0]: B			0	0	Pixel[0,0]: B

5.5. 16-bit RGB Planar

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
RGB48PL	N/A	PLANAR	6 Bytes/pixel

Memory Layout - Plane 0

@	+3	+2	+1	+0
0	Pixel[1,0]: R		Pixel[0,0]: R	

Memory Layout - Plane 1

@	+3	+2	+1	+0
0	Pixel[1,0]: G		Pixel[0,0]: G	

Memory Layout - Plane 2

@	+3	+2	+1	+0
0	Pixel[1,0]: B		Pixel[0,0]: B	

6. YUV Color Pixel Formats

MultiCam Name	PFNC Name	Link
YUV411 Y41P	YUV411_8_UYVYUYVYYYYY	"8-bit YUV 4:1:1" on the next page
YUV422 Y42P	YUV422_8_YUYV	"8-bit YUV 4:2:2" on page 38
YUV444 IYU2	YUV444_8_UYV	"8-bit YUV 4:4:4" on page 39

6.1. 8-bit YUV 4:1:1

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
YUV411	YUV411_8_ UYVYUYVYYYYY	PACKED	1.5 Bytes/pixel
Y41P			

Memory Layout

@	+3	+2	+1	+0
0	Pixel[1,0]: Y	Pixel[0,0]: V(Cr)	Pixel[0,0]: Y	Pixel[0,0]: U(Cb)
4	Pixel[3,0]: Y	Pixel[4,0]: V(Cr)	Pixel[2,0]: Y	Pixel[4,0]: U(Cb)
8	Pixel[7,0]: Y	Pixel[6,0]: Y	Pixel[5,0]: Y	Pixel[4,0]: Y



NOTE

Only 1 U(Cb) and 1 V(Cr) samples are captured every 4 pixels

6.2. 8-bit YUV 4:2:2

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
YUV422	YUV422_8_YUYV	PACKED	2 Bytes/pixel
Y42P			

Memory Layout

@	+3	+2	+1	+0
0	Pixel[0,0]: V(Cr)	Pixel[1,0]: Y	Pixel[0,0]: U(Cb)	Pixel[0,0]: Y


NOTE

Only 1 U(Cb) and 1 V(Cr) samples are captured every 2 pixels

6.3. 8-bit YUV 4:4:4

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
YUV444	YUV444_8_UYV	PACKED	3 Bytes/pixel
IYU2			

Memory Layout

@	+3	+2	+1	+0
0	Pixel[1,0]: U(Cb)	Pixel[0,0]: V(Cr)	Pixel[0,0]: Y	Pixel[0,0]: U(Cb)
4	Pixel[2,0]: Y	Pixel[2,0]: U(Cb)	Pixel[1,0]: V(Cr)	Pixel[1,0]: Y
8	Pixel[3,0]: V(Cr)	Pixel[3,0]: Y	Pixel[3,0]: U(Cb)	Pixel[2,0]: V(Cr)

7. YUV Color Planar Pixel Formats

MultiCam Name	PFNC Name	Link
YUV411PL_Dec YUV9 YVU9	YUV410_8_Planar	"8-bit YUV 4:1:0 Planar" on the next page
YUV411PL Y41B	YUV411_8_Planar	"8-bit YUV 4:1:1 Planar" on page 43
YUV422PL_Dec I420 IYUV YV12	YUV420_8_Planar	"8-bit YUV 4:2:0 Planar" on page 44
YUV422PL Y42B	YUV422_8_Planar	"8-bit YUV 4:2:2 Planar" on page 46
YUV444PL	YUV444_8_Planar	"8-bit YUV 4:4:4 Planar" on page 47

7.1. 8-bit YUV 4:1:0 Planar

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
YUV411PL_Dec	YUV410_8_Planar	PLANAR	1.125 Bytes/pixel
YUV9			
YVU9			

Memory Layout - Plane 0

@	+3	+2	+1	+0
0	Pixel[3,0]: Y	Pixel[2,0]: Y	Pixel[1,0]: Y	Pixel[0,0]: Y


NOTE

H = buffer pitch (in bytes)

Memory Layout - Plane 1

@	+3	+2	+1	+0
0	Pixel[12,0]: U(Cb)	Pixel[8,0]: U(Cb)	Pixel[4,0]: U(Cb)	Pixel[0,0]: U(Cb)
:	:	:	:	:
H	Pixel[12,4]: U(Cb)	Pixel[8,4]: U(Cb)	Pixel[4,4]: U(Cb)	Pixel[0,4]: U(Cb)
:	:	:	:	:


NOTE

Only 1 U(Cb) sample is captured every 4 pixels in 1 line every 4 lines

Memory Layout - Plane 2

@	+3	+2	+1	+0
0	Pixel[12,0]: V(Cr)	Pixel[8,0]: V(Cr)	Pixel[4,0]: V(Cr)	Pixel[0,0]: V(Cr)
:	:	:	:	:
0	Pixel[12,4]: V(Cr)	Pixel[8,4]: V(Cr)	Pixel[4,4]: V(Cr)	Pixel[0,4]: V(Cr)
:	:	:	:	:

**NOTE**

Only 1 V(Cr) sample is captured every 4 pixels in 1 line every 4 lines

7.2. 8-bit YUV 4:1:1 Planar

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
YUV411PL	YUV411_8_Planar	PLANAR	1.5 Bytes/pixel
Y41B			

Memory Layout - Plane 0

@	+3	+2	+1	+0
0	Pixel[3,0]: Y	Pixel[2,0]: Y	Pixel[1,0]: Y	Pixel[0,0]: Y
4	Pixel[7,0]: Y	Pixel[6,0]: Y	Pixel[5,0]: Y	Pixel[4,0]: Y
8	Pixel[11,0]: Y	Pixel[10,0]: Y	Pixel[9,0]: Y	Pixel[8,0]: Y
12	Pixel[15,0]: Y	Pixel[14,0]: Y	Pixel[13,0]: Y	Pixel[12,0]: Y

Memory Layout - Plane 1

@	+3	+2	+1	+0
0	Pixel[12,0]: U(Cb)	Pixel[8,0]: U(Cb)	Pixel[4,0]: U(Cb)	Pixel[0,0]: U(Cb)


NOTE

Only 1 U(Cb) sample is captured every 4 pixels

Memory Layout - Plane 2

@	+3	+2	+1	+0
0	Pixel[12,0]: V(Cr)	Pixel[8,0]: V(Cr)	Pixel[4,0]: V(Cr)	Pixel[0,0]: V(Cr)


NOTE

Only 1 V(Cr) sample is captured every 4 pixels

7.3. 8-bit YUV 4:2:0 Planar

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
YUV422PL_Dec	YUV420_8_Planar	PLANAR	1.5 Bytes/pixel
I420			
IYUV			
YV12			

Memory Layout - Plane 0

@	+3	+2	+1	+0
0	Pixel[3,0]: Y	Pixel[2,0]: Y	Pixel[1,0]: Y	Pixel[0,0]: Y

Memory Layout - Plane 1

@	+3	+2	+1	+0
0	Pixel[6,0]: U(Cb)	Pixel[4,0]: U(Cb)	Pixel[2,0]: U(Cb)	Pixel[0,0]: U(Cb)
:	:	:	:	:
2 H	Pixel[6,2]: U(Cb)	Pixel[4,2]: U(Cb)	Pixel[2,2]: U(Cb)	Pixel[0,2]: U(Cb)
:	:	:	:	:


NOTE

Only 1 U(Cb) sample is captured every 2 pixels in 1 line every 2 lines

Memory Layout - Plane 2

@	+3	+2	+1	+0
0	Pixel[6,0]: V(Cr)	Pixel[4,0]: V(Cr)	Pixel[2,0]: V(Cr)	Pixel[0,0]: V(Cr)
:	:	:	:	:
H	Pixel[6,2]: V(Cr)	Pixel[4,2]: V(Cr)	Pixel[2,2]: V(Cr)	Pixel[0,2]: V(Cr)
:	:	:	:	:

**NOTE**

Only 1 V(Cr) sample is captured every 2 pixels in 1 line every 2 lines

7.4. 8-bit YUV 4:2:2 Planar

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
YUV422PL	YUV422_8_Planar	PLANAR	2 Bytes/pixel
Y42B			

Memory Layout - Plane 0

@	+3	+2	+1	+0
0	Pixel[3,0]: Y	Pixel[2,0]: Y	Pixel[1,0]: Y	Pixel[0,0]: Y
4	Pixel[7,0]: Y	Pixel[6,0]: Y	Pixel[5,0]: Y	Pixel[4,0]: Y

Memory Layout - Plane 1

@	+3	+2	+1	+0
0	Pixel[6,0]: U(Cb)	Pixel[4,0]: U(Cb)	Pixel[2,0]: U(Cb)	Pixel[0,0]: U(Cb)


NOTE

Only 1 U(Cb) sample is captured every 2 pixels

Memory Layout - Plane 2

@	+3	+2	+1	+0
0	Pixel[6,0]: V(Cr)	Pixel[4,0]: V(Cr)	Pixel[2,0]: V(Cr)	Pixel[0,0]: V(Cr)


NOTE

Only 1 V(Cr) sample is captured every 2 pixels

7.5. 8-bit YUV 4:4:4 Planar

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
YUV444PL	YUV444_8_Planar	PLANAR	3 Bytes/pixel

Memory Layout - Plane 0

@	+3	+2	+1	+0
0	Pixel[3,0]: Y	Pixel[2,0]: Y	Pixel[1,0]: Y	Pixel[0,0]: Y

Memory Layout - Plane 1

@	+3	+2	+1	+0
0	Pixel[3,0]: U(Cb)	Pixel[2,0]: U(Cb)	Pixel[1,0]: U(Cb)	Pixel[0,0]: U(Cb)

Memory Layout - Plane 2

@	+3	+2	+1	+0
0	Pixel[3,0]: V(Cr)	Pixel[2,0]: V(Cr)	Pixel[1,0]: V(Cr)	Pixel[0,0]: V(Cr)

8. Raw Data Formats

MultiCam Name	PFNC Name	Link
RAW8	Raw8	"8-bit Raw Data" on the next page
RAW10	Raw10	"10-bit Raw Data" on page 50
RAW12	Raw12	"12-bit Raw Data" on page 51
RAW14	Raw14	"14-bit Raw Data" on page 52
RAW16	Raw16	"16-bit Raw Data" on page 53

8.1. 8-bit Raw Data

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8.2. 10-bit Raw Data

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8.3. 12-bit Raw Data

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8.4. 14-bit Raw Data

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8.5. 16-bit Raw Data

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8.1. 8-bit Raw Data

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
RAW8	Raw8	N/A	1 Byte/pixel

Memory Layout

@	+3	+2	+1	+0
0	Pixel[3,0]	Pixel[2,0]	Pixel[1,0]	Pixel[0,0]

8.2. 10-bit Raw Data

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
RAW10	Raw10	N/A	2 Bytes/pixel

Memory Layout

@	+3							+2							+1							+0						
0	0	0	0	0	0	0	0	Pixel[1,0]							0	0	0	0	0	0	0	Pixel[0,0]						

8.3. 12-bit Raw Data

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
RAW12	Raw12	N/A	2 Bytes/pixel

Memory Layout

@	+3					+2	+1				+0
0	0	0	0	0	0	Pixel[1,0]	0	0	0	0	Pixel[0,0]

8.4. 14-bit Raw Data

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
RAW14	Raw14	N/A	2 Bytes/pixel

Memory Layout

@	+3			+2	+1			+0
0	0	0	Pixel[1,0]			0	0	Pixel[0,0]

8.5. 16-bit Raw Data

MultiCam Name	PFNC Name	Storage Type	Storage Requirement
RAW16	Raw16	N/A	2 Bytes/pixel

Memory Layout

@	+3	+2	+1	+0
0	Pixel[1,0]		Pixel[0,0]	